

Force Control

Control of Robot Manipulators by F.L.Lewis, C.T.Abdalla, D.M.Dawson
Robotics: modelling, planning and control by B. Siciliano, L. Sciavicco,
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From Previous Lecture

- Derivation of Control

- System: $\tau = M(\Theta)\ddot{\Theta} + V(\Theta, \dot{\Theta}) + G(\Theta) + F(\Theta, \dot{\Theta})$

- Our control: $\tau = \alpha\tau' + \beta$

- Closed loop system

- Our control law: $\tau = \hat{M}(\ddot{\Theta}_d + K_v\dot{\Theta} + K_p\Theta) + \hat{V} + \hat{G} + \hat{F}$

- Then,

- $M\ddot{\Theta} + V + G + F = \hat{M}(\ddot{\Theta}_d + K_v\dot{\Theta} + K_p\Theta) + \hat{V} + \hat{G} + \hat{F}$

- $M\ddot{\Theta} + \hat{M}\ddot{\Theta} - \hat{M}\ddot{\Theta} + V + G + F = \hat{M}(\ddot{\Theta}_d + K_v\dot{\Theta} + K_p\Theta) + \hat{V} + \hat{G} + \hat{F}$

- $\hat{M}(\ddot{\Theta}_d - \ddot{\Theta} + K_v\dot{\Theta} + K_p\Theta) = (M - \hat{M})\ddot{\Theta} + (V - \hat{V}) + (G - \hat{G}) + (F - \hat{F})$

- $\ddot{E} + K_v\dot{\Theta} + K_p\Theta = \hat{M}^{-1}[(M - \hat{M})\ddot{\Theta} + (V - \hat{V}) + (G - \hat{G}) + (F - \hat{F})] \approx \hat{M}^{-1}\tau_d$

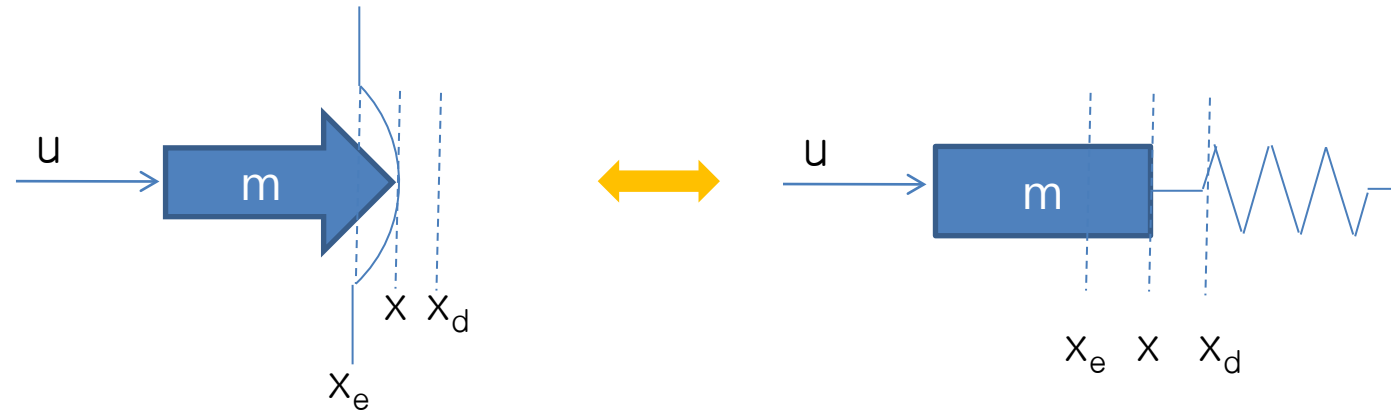
$$\begin{aligned}\alpha &= \hat{M}(\Theta), \\ \beta &= \hat{V} + \hat{G} + \hat{F} \\ \tau' &= \ddot{\Theta}_d + K_v\dot{\Theta} + K_p\Theta\end{aligned}$$

Agenda

- Indirect force control
 - Stiffness control – operational space control
 - Impedance control
- Direct force control
 - Hybrid force/motion control

Stiffness Control (Motivation)

- The force exerted on the environment : $f = k_e(x - x_e)$



- EOM of the equivalent system:
- Choose PD control ($\dot{x}_d = 0$):

$$m\ddot{x} + k_e(x - x_e) = u$$

$$u = k_v(\dot{x}_d - \dot{x}) + k_p(x_d - x) = -k_v\dot{x} + k_p(x_d - x)$$

- Closed loop equation:

$$m\ddot{x} + k_v\dot{x} + (k_p + k_e)x = k_px_d + k_ex_e$$

Stiffness Control (Motivation)

- Closed loop system $m\ddot{x} + k_v\dot{x} + (k_p + k_e)x = k_px_d + k_ex_e$
- Steady-state position ($\dot{x} \rightarrow 0, \ddot{x} \rightarrow 0$): $\bar{x} = \frac{k_px_d + k_ex_e}{k_p + k_e}$
- Assumption: environment is near rigid ($k_e \gg k_p$)
- Steady-state force:
 - $\bar{k} = k_e(\bar{x} - x_e) = k_e \left(\frac{k_px_d + k_ex_e}{k_p + k_e} - x_e \right) = k_e \left(\frac{k_p(x_d - x_e)}{k_p + k_e} \right) \approx k_p(x_d - x_e)$

PD control exerts a spring-like-force on the environment

$$u = -k_v\dot{x} + k_p(x_d - x)$$

Jacobians in the Force Domain

- Virtual work analysis
 - The work done in Cartesian space = work done in joint-space

$$F \cdot \delta x = \tau \cdot \delta \theta \quad \text{or} \quad F^T \delta x = \tau^T \delta \theta$$

- Jacobian: $\delta x = J \delta \theta$
- Thus, $F^T J \delta \theta = \tau^T \delta \theta$
- This must hold for all variation of the joint angle

- Recall:

$$\tau = J^T F$$

$$\dot{x} = J \dot{\theta}$$

Stiffness Control of N-link Manipulator

$$M\ddot{q} + V_m\dot{q} + G(q) + F(\dot{q}) + J^T \overset{\text{Contact force}}{f} = \tau$$

- Environmental surface friction is neglected
- If manipulator is not constrained in a particular task space direction, the corresponding K_e diagonal element is zero
- Force exerted on the environment

- Stiffness controller $f = K_e(x - x_e)$
- Then, closed loop becomes; $\tau = J^T(q)(-K_v\dot{x} + K_p\tilde{x}) + G(q) + F(\dot{q})$
 $\tilde{x} = x_d - x$

- Now, we need to study the stability of the closed system

$$M\ddot{q} + V_m\dot{q} = J^T(-K_v\dot{x} + K_p\tilde{x} - K_e(x - x_e))$$

Stability of Stiffness Control

- Consider Lyapunov candidate

$$V = \frac{1}{2} \dot{q}^T M(q) \dot{q} + \frac{1}{2} \tilde{x}^T K_p \tilde{x} + \frac{1}{2} (x - x_e)^T K_e (x - x_e)$$

- Now, $\dot{V} = \frac{1}{2} \dot{q}^T \dot{M} \dot{q} + \dot{q}^T \boxed{M \ddot{q}} - \dot{q}^T J^T K_p \tilde{x} + \dot{q}^T J^T K_e (x - x_e)$

$$M \ddot{q} + V_m \dot{q} = J^T (-K_v \dot{x} + K_p \tilde{x} - K_e (x - x_e))$$

- Together with skew-symmetric property,

$$\dot{V} = -\dot{q}^T J^T K_v J \dot{q} \leq 0$$

- Steady-state position and force

$$\lim_{t \rightarrow \infty} x_i = (K_{pi} + K_{ei})^{-1} (K_{pi} x_{di} + K_{ei} x_{ei})$$

$$\lim_{t \rightarrow \infty} f_i = K_{ei} (K_{pi} + K_{ei})^{-1} (K_{pi} x_{di} + K_{ei} x_{ei})$$

- If $K_e \gg K_p$ (environment is more stiff than robot)

$$\lim_{t \rightarrow \infty} f_i \approx K_{pi} (x_{di} - x_{ei})$$

Hybrid Position/Force Control

- Disadvantage of stiffness control: set-point control
- Separate position and force controller is desired !!!
- Toy problem: Cartesian Two-link Robot

$$x = \begin{bmatrix} u \\ v \end{bmatrix} = h(q) = \begin{bmatrix} q_1 \\ q_2 \end{bmatrix}$$

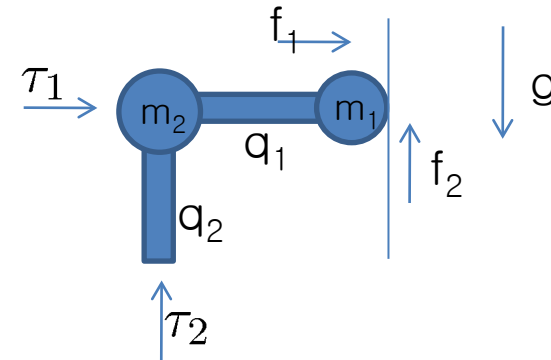
$$J = \frac{\partial h(q)}{\partial q} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

- Manipulator dynamics

$$\tau = M\ddot{q} + G + f$$

$$\tau_1 = m_1\ddot{q}_1 + f_1, \quad \tau_2 = (m_1 + m_2)\ddot{q}_2 + (m_1 + m_2)g + f_2$$

- Position control in q_2 , force control in q_1 directions



Hybrid P/F Control: Toy Problem

- Computed Torque Control for position

$$\tau_2 = (m_1 + m_2)(\ddot{q}_{d2} + k_{Tv}\dot{\tilde{x}} + k_{Tp}\tilde{x}) + (m_1 + m_2)g + f_2$$

$$(m_1 + m_2)\ddot{q}_2 + (m_1 + m_2)g + f_2 = \tau_2$$

$$\tilde{x} = q_{d2} - q_2$$

- Closed loop system: $\ddot{\tilde{x}} + k_{Tv}\dot{\tilde{x}} + k_{Tp}\tilde{x} = 0$
- Position controller requires measurement of the joint position, joint velocity, and surface friction force

Force Control using CTM

- Model the environment as a spring:

$$f_1 = k_e(q_1 - q_e) \quad \longrightarrow \quad \ddot{q}_1 = \frac{1}{k_e} \ddot{f}_1$$

Normal force exerted on the environment

- Force dynamic equation:

$$\tau_1 = \frac{m_1}{k_e} \ddot{f}_1 + f_1$$

- CTM for force control:

$$\tau_1 = \frac{m_1}{k_e} (\ddot{f}_{d1} + k_{Nv} \dot{\tilde{f}} + k_{Np} \tilde{f}) + f_1$$

$$\tilde{f} = f_{d1} - f_1$$

- Closed loop system:

$$\ddot{\tilde{f}} + k_{Tv} \dot{\tilde{f}} + k_{Tp} \tilde{f} = 0$$

- Force controller requires measurement of the normal force and the derivative of the normal force → use

$$\dot{f}_1 = k_e \dot{q}_1$$

Hybrid P/F Control of N-link Arm

- Feedback linearization

$$\tau = M(q)J^{-1}(q)(\mathbf{a} - \dot{J}(q)\dot{q}) + V_m(q, \dot{q})\dot{q} + G(q) + F(\dot{q}) + J^T(q)f$$

- Position control

$$\mathbf{a}_{T_i} = \ddot{x}_{T_{d_i}} + k_{T_{v_i}}\dot{\tilde{x}}_{T_{v_i}} + k_{T_{p_i}}\tilde{x}_{T_{p_i}}$$

- Force control

$$\mathbf{a}_{N_j} = \frac{1}{k_{e_j}} \left(\ddot{f}_{N_{d_j}} + k_{N_{v_j}}\dot{\tilde{f}}_{N_{v_j}} + k_{N_{p_j}}\tilde{f}_{N_{p_j}} \right)$$

- Implementation issues

- Task space forces are needed
- Sensors are mounted in the wrist → transformation

$$\tau = J^T(q)f = J_s^T(q)f_s \Rightarrow f = J^{-T}(q)J_s^T(q)f_s$$

- Task space force derivative

$$\dot{\tilde{f}}_{N_j} = k_{e_j}\dot{\tilde{x}}_{N_j}$$

Hybrid Impedance Control

- Modeling the environment

- Impedance model

$$f(s) = Z_e(s)\dot{x}(s)$$

Force exerted on the environment

Environmental impedance

Velocity of the manipulator at the contact point

- Spring-damper model

$$f = h\ddot{x} + b\dot{x} + kx$$

- Capacitive environmental impedance

$$Z_e(s) = hs + b + \frac{k}{s}$$

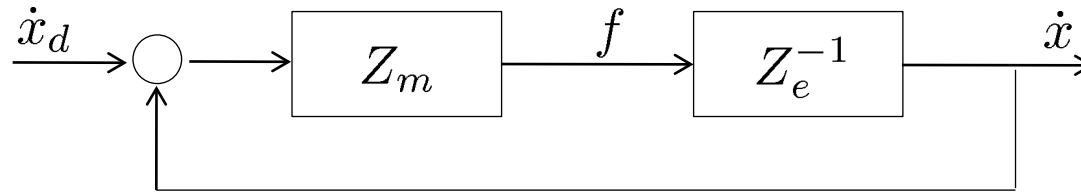
- How does one design a controller for a given environmental impedance?

- Choose a proper manipulator (target) impedance

Position and Force Control Models

- Position control

$$f(s) = Z_m(s)(\dot{x}_d(s) - \dot{x}(s))$$



- For step input,

$$E_{ss} = \lim_{s \rightarrow 0} s(\dot{x}_d(s) - \dot{x}(s)) = \lim_{s \rightarrow 0} \frac{Z_e(s)}{Z_m(s) + Z_e(s)} \Rightarrow 0$$

Inertial impedance

when $|Z_e(0)| = c$, or $|Z_e(0)| = 0$

and $|Z_m(0)| = c$, or $|Z_m(0)| = \infty$

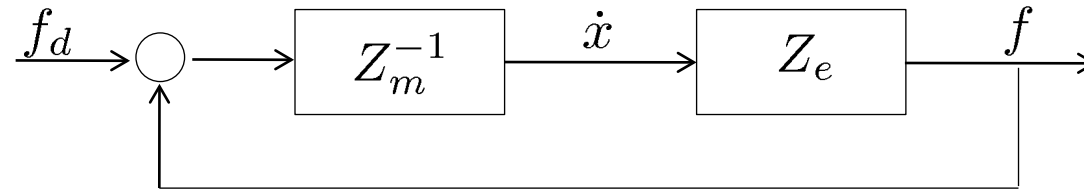
Resistive impedance Capacitive impedance

- Zero steady-state error for a velocity step input if
 - Inertial environments are position controlled with non-inertial manipulator impedance, while resistive environments are position controlled with capacitive manipulator impedance

Force Control: Duality

- Force control:

$$\dot{x}(s) = Z_m^{-1}(s)(f_d(s) - f(s))$$



- For step input,

$$E_{ss} = \lim_{s \rightarrow 0} s(f_d(s) - f(s)) = \lim_{s \rightarrow 0} \frac{Z_m(s)}{Z_m(s) + Z_e(s)} \Rightarrow 0$$

Capacitive impedance

when $|Z_e(0)| = c$, or $|Z_e(0)| = \infty$

and $\underbrace{|Z_m(0)| = c}_{\text{Resistive impedance}}$, or $\underbrace{|Z_m(0)| = 0}_{\text{Inertial impedance}}$

Duality Principle

- Capacitive environments are force controlled with noncapacitive manipulator impedance
- Inertial environments are position controlled with noninertial manipulator impedances
- Resistive environments are force controlled with inertial manipulator impedance or position controlled with capacitive manipulator impedance

Impedance Control Formulation

- Torque control for feedback linearization

$$\tau = M(q)J^{-1}(q)(\mathbf{a} - \dot{J}(q)\dot{q}) + V_m(q, \dot{q})\dot{q} + G(q) + F(\dot{q}) + J^T(q)f$$

- Linear system: $\ddot{x} = \mathbf{a}$ ← Control strategies

- Task space is globally linearized
- Separate position and force controllers

- Position Control $\ddot{x}_{pi} = \mathbf{a}_{pi}$

- Zero initial value: $s\dot{x}_{pi}(s) = \mathbf{a}_{pi}(s) = s(\dot{x}_{pdi}(s) - Z_{pmi}^{-1}(s)f_{pi}(s))$

$$\mathbf{a}_{pi}(s) = L^{-1} \left\{ s(\dot{x}_{pdi}(s) - Z_{pmi}^{-1}(s)f_{pi}(s)) \right\}$$

- Force Control $\ddot{x}_{fj} = \mathbf{a}_{fj}$

- Zero initial value: $s\dot{x}_{fj}(s) = \mathbf{a}_{fj}(s) = sZ_{fmj}^{-1}(s)(f_{fdj}(s) - f_{fj}(s))$

$$\mathbf{a}_{fj}(s) = L^{-1} \left\{ sZ_{fmj}^{-1}(s)(f_{fdj}(s) - f_{fj}(s)) \right\}$$

Implementation Issues

- For practical reasons, manipulator impedance should not contain acceleration or force derivative
 - A candidate: $Z_m = h\dot{s} + Z_r$
- Task space forces should be represented by the sensor forces

$$\tau = J^T(q)f = J_s^T(q)f_s \Rightarrow f = J^{-T}(q)J_s^T(q)f_s$$

Cartesian Impedance Control for DLR

H. Liu and G. Hirzinger, Cartesian Impedance Control for the DLR Hand, IROS 1999

- Kinematic modeling

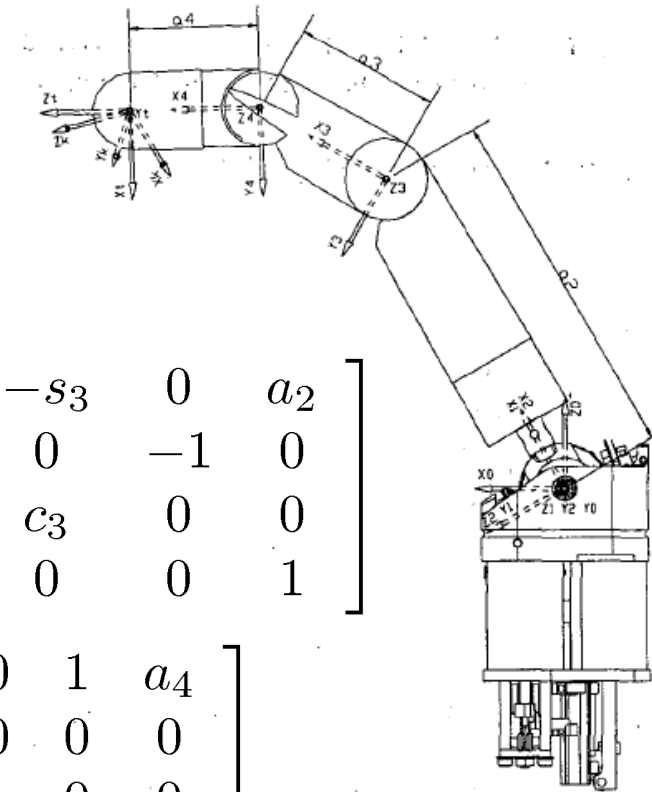
$${}^0_1T = \begin{bmatrix} s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ c_1 & -s_1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1_2T = \begin{bmatrix} c_2 & -s_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -s_2 & -c_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^3_4T = \begin{bmatrix} c_4 & -s_4 & 0 & a_3 \\ s_4 & c_4 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2_3T = \begin{bmatrix} c_3 & -s_3 & 0 & a_2 \\ 0 & 0 & -1 & 0 \\ s_3 & c_3 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^4_{\text{Tip}}T = \begin{bmatrix} 0 & 0 & 1 & a_4 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Cartesian Impedance Control

- Jacobian

$$J = \begin{bmatrix} c_2(a_2c_{34} + a_3c_4 + a_4) & 0 & a_3c_4 + a_4 & a_4 \\ -s_2(a_3s_3 + a_4s_{34}) & -(a_2 + a_3c_3 + a_4c_{34}) & 0 & 0 \\ c_2(a_2s_{34} + a_3s_4) & 0 & a_3s_4 & 0 \end{bmatrix}$$

- Complete impedance of a finger

$$F_{\text{ext}} = M_d \ddot{x}_k + B_d \delta \dot{x}_k + K_d \delta x_k$$

- Fingertip coordinate Position error $\delta x_k = x - x_d$

$$\tau_{\text{ext}} = J_4^T F_{\text{ext}}$$

Jacobian of two base joints and the last joint

$$J_4 = \begin{bmatrix} c_2(a_2c_{34} + a_3c_4 + a_4) & 0 & a_4 \\ -s_2(a_3s_3 + a_4s_{34}) & -(a_2 + a_3c_3 + a_4c_{34}) & 0 \\ c_2(a_2s_{34} + a_3s_4) & 0 & 0 \end{bmatrix}$$

Cartesian Impedance Control

- Cartesian displacement $\delta x_k \approx \tilde{J} \delta \theta$
- Coupling between 3rd link and 4th link

$$\tilde{J} = \begin{bmatrix} c_2(a_2c_{34} + a_3c_4 + a_4) & 0 & a_3c_4 + 2a_4 \\ -s_2(a_3s_3 + a_4s_{34}) & -(a_2 + a_3c_3 + a_4c_{34}) & 0 \\ c_2(a_2s_{34} + a_3s_4) & 0 & a_3s_4 \end{bmatrix}$$

- Now,

$$\begin{aligned} \tau_{\text{ext}} &= J_4^T (M_d \ddot{x}_k + B_d \delta \dot{x}_k + K_d \delta x) \\ &= J_4^T \left(M_d (\tilde{J} \delta \dot{\theta} + \tilde{J} \delta \ddot{\theta}) + B_d \tilde{J} \delta \dot{\theta} + K_d \tilde{J} \delta \theta \right) \\ &= M_\theta \delta \ddot{\theta} + B_\theta \delta \dot{\theta} + K_\theta \delta \theta \\ M_\theta &= J_4^T M_d \tilde{J}, \quad B_\theta = J_4^T M_d \dot{\tilde{J}} + J_4^T B_d \tilde{J}, \quad K_\theta = J_4^T K_d \tilde{J} \end{aligned}$$

- Arbitrary coordination
 - Consider the rotational matrix between the new Cartesian impedance system and the fingertip one

$$M'_\theta = J_4^T {}^t_k R M_d {}^k_t R \tilde{J}, \quad B_\theta = J_4^T {}^t_k R M_d {}^k_t R \dot{\tilde{J}} + J_4^T {}^t_k R B_d {}^k_t R \tilde{J}, \quad K_\theta = J_4^T {}^t_k R K_d {}^k_t R \tilde{J}$$

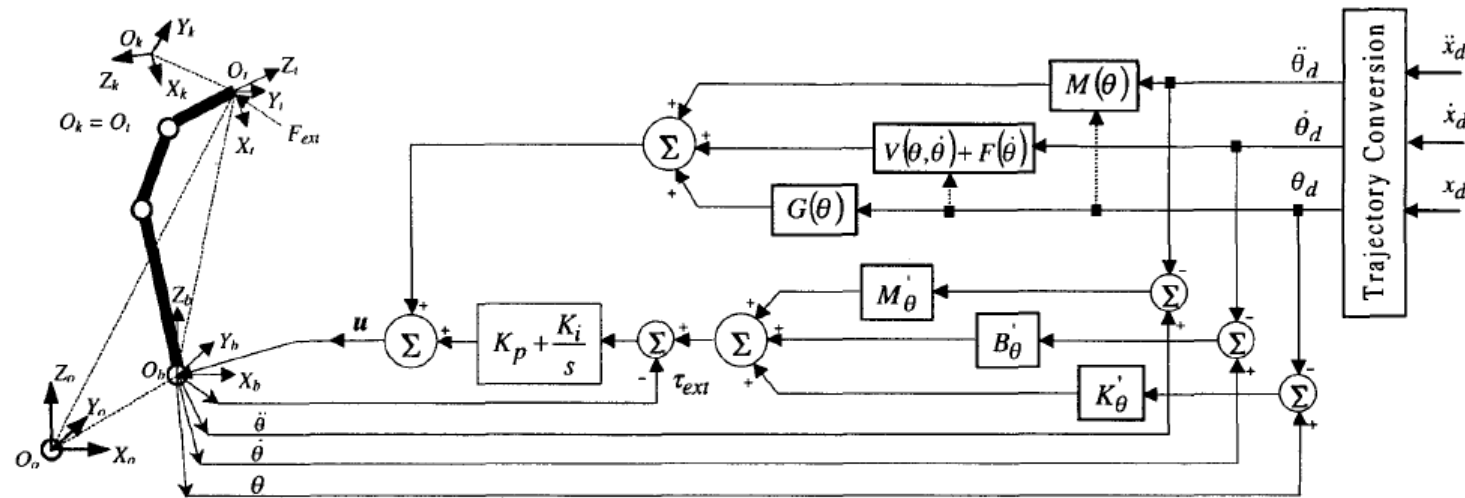
Implementation

- Feedback linearization control

$$\tau_{\text{act}} = M(\theta) \left(\ddot{\theta} + M_{\theta}^{-1} \left(\tau_{\text{ext}} - B_{\theta} \delta \dot{\theta} - K_{\theta} \delta \theta \right) + V(\theta, \dot{\theta}) + G(\theta) + F(\dot{\theta}, \text{sign}(\dot{\theta})) + J_4^T F_{\text{ext}} \right)$$

- Outer loop PI control for impedance error

$$\tau_e = \tau_d - \tau_{\text{ext}} \quad \tau_d = M_{\theta} \delta \ddot{\theta} + B_{\theta} \delta \dot{\theta} + K_{\theta} \delta \theta$$



$O_0 - X_0 Y_0 Z_0$: Hand Coordinate System
 $O_b - X_b Y_b Z_b$: Finger Base Coordinate System
 $O_t - X_t Y_t Z_t$: Fingertip Coordinate System
 $O_k - X_k Y_k Z_k$: Desired Impedance Coordinate System

$M_{\theta} = J_{A_k}^T R M_{d_k}^k R J$
 $B_{\theta} = J_{A_k}^T R M_{d_k}^k R \dot{J} + J_{A_k}^T R B_{d_k}^k R J$
 $K_{\theta} = J_{A_k}^T R K_{d_k}^k R J$

M_d : Desired Inertia in $O_k - X_k Y_k Z_k$
 B_d : Desired Damping in $O_k - X_k Y_k Z_k$
 K_d : Desired Stiffness in $O_k - X_k Y_k Z_k$